GITAM Deemed to be University

Civil Engineering | Course Code: CE304

Transportation & Highway Engineering - Highway Geometric Design & Pavement

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Overview: Stopping Sight Distance SSD, Overtaking Sight Distance OSD, Super-elevation design, C

I. UNIT-WISE SYLLABUS BREAKDOWN & HIGH-YIELD TOPICS

Unit 1: Highway Planning & Alignments
Highway development plans, Road patterns, Engineering surveys for highway location, Environmental im

Unit 2: Geometric Design of Highways
Design speed, Cross-sectional elements, Camber, Sight distances SSD, OSD, Horizontal alignment, Supe

Unit 3: Pavement Materials & Testing
Subgrade soil characterization, CBR test, Crushing and Impact value of aggregates, Bitumen penetrati

Unit 4: Pavement Design Principles
Design of flexible pavements as per IRC:37 guidelines, Westergaard stress equations for rigid paveme

II. HIGH-YIELD SEMESTER EXAMINATION QUESTIONS

1. Calculate the design super-elevation and length of transition curve for a highway curve of radius 400
2. Describe the California Bearing Ratio CBR laboratory testing procedure and its application in flexible
3. Compute the safe stopping sight distance SSD on a highway with 70 km/h design speed on a 3% downgrade

III. PRESCRIBED TEXTBOOKS & ACADEMIC REPOSITORIES

* S.K. Khanna, C.E.G. Justo, A. Veeraragavan - Highway Engineering Nem Chand & Bros 10th Ed
* Paul H. Wright, Karen Dixon - Highway Engineering Wiley 7th Ed

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